F70LA Life Insurance Mathematics A (2025/26)
Tutorial 3

1. On the basis of ELT15 (Females) mortality, evaluate the following:

   (i) ${}_{14}p_{37}$, (ii) ${}_{8}q_{60}$.

2. A mortality table has a limiting age of 100 years and is such that for $0 \leq x \leq 100$,

$$l_x = 1,000,000\sqrt{100 - x}.$$

   Calculate the complete expectation of life at exact age 60.

3. Using English Life Table Number 15 - Males, find the probability that a man aged 30 will

   (i) Survive to age 40, (ii) Die before reaching age 50,
   (iii) Die in his 45th year of life.

4. Calculate ${}_{20}q_{65}$ on the mortality basis of

   (i) ELT15 (Males), (ii) AM92 Ultimate,
   (iii) PMA92C20.

5. Calculate ${}_{8}|_{7}q_{62}$ on the mortality basis of

   (i) ELT15 (Females), (ii) AM92 Ultimate,
   (iii) PFA92C20.

6. (a) State the Uniform Distribution of Deaths (UDD) assumption between integer ages.

   (b) Show that the UDD assumption implies that for integer x and for $0 \leq s, t \leq 1$ such that $s + t \leq 1$,

$$ {}_{t}q_{x+s} = \frac{t \cdot q_x}{1 - s \cdot q_x}. $$

   (c) Evaluate ${}_{0.75}q_{60.75}$ on the basis of PMA92C20 mortality, assuming UDD over integer ages.

7. On the basis of ELT15 (Males) mortality, calculate the values of

   (i) $l_{56.6}$, (ii) ${}_{0.4}q_{60}$, (iii) ${}_{3}p_{45.25}$, (iv) $\mu_{64.2}$,

   assuming a uniform distribution of deaths between integer ages.

8. Using ELT15 Males mortality, assume that between exact ages 60 and 62, l_x is a quadratic function of x. Using the values of $l_{60} = 86714$, $l_{61} = 85507$ and $l_{62} = 84173$, derive an expression for l_{60+t} which holds for $0 \leq t \leq 2$.

   Hence calculate the value of ${}_{0.4}q_{60}$.

9. The table below shows the expected number of people alive at age 75 upwards, based on a select period of 2 years.

x	$\ell_{[x]}$	$\ell_{[x]+1}$	ℓ_{x+2}	$x+2$
75	15 930	15 668	15 286	77
76	15 508	15 224	14 816	78
77	15 050	14 744	14 310	79
⋮	⋮	⋮	⋮	⋮
80			12 576	82
81			11 928	83
82			11 250	84
83			10 542	85
84			9 812	86
85			9 064	87

   Using the table, calculate the following:

   (a) The probability that a life currently age 75, who has just been selected, will survive to age 85.

   (b) The probability that a life currently age 76, who was selected one year ago, will die between ages 85 and 87.

   (c) ${}_{4|2}q_{[77]+1}$.

10. The table below is based on UK data for female non-smokers who are term insurance policyholders. It has a select period of five years. The table shows the values of $q_{[x-t]+t}$, for $t \in \{0, 1, 2, 3, 4, 5\}$

Age x	Duration 0 $q_{[x]}$	Duration 1 $q_{[x-1]+1}$	Duration 2 $q_{[x-2]+2}$	Duration 3 $q_{[x-3]+3}$	Duration 4 $q_{[x-4]+4}$	Durations 5+ q_x
69	0.003974	0.004979	0.005984	0.006989	0.007994	0.009458
70	0.004285	0.005411	0.006537	0.007663	0.008790	0.010599
71	0.004704	0.005967	0.007229	0.008491	0.009754	0.011880
72	0.005236	0.006651	0.008066	0.009481	0.010896	0.013318
73	0.005870	0.007456	0.009043	0.010629	0.012216	0.014931
74	0.006582	0.008361	0.010140	0.011919	0.013698	0.016742
75	0.007381	0.009376	0.011370	0.013365	0.015360	0.018774
76	0.008277	0.010514	0.012751	0.014988	0.017225	0.021053
77	0.009281	0.011790	0.014299	0.016807	0.019316	0.023609

Use the table to calculate the following probabilities:

(a) ${}_2p_{[72]}$; (b) ${}_3q_{[73]+2}$; (c) ${}_1|q_{[65]+4}$; (d) ${}_7p_{[70]}$.

Numerical solutions: 1(a) 0.97706; 1(b) 0.09356. Q2 26.67 years. Q3(a) 0.98827; 3(b) 0.038097; 3(c) 0.002355. Q4(a) 0.75085; 4(b) 0.6162399; 4(c) 0.49285. Q5(a) 0.17096; 5(b) 0.1939900; 5(c) 0.098874. Q6(c) 0.00209. Q7(i) 90006.4; 7(ii) 0.005568; 7(iii) 0.9908100; 7(iv) 0.022087; Q8 0.0053920; Q9(a) 0.66177; Q9(b) 0.09433; Q9(c) 0.08993; Q10(a) 0.98735; Q10(b) 0.04500; Q10(c) 0.01051; Q10(d) 0.92027.

ELT15 (Females)

x	l_x	d_x	q_x	μ_x	$\mathring{e}_x$	x
0	100 000	632	0.006 32		78.956	0
1	99 368	55	0.000 55	0.000 73	78.462	1
2	99 313	30	0.000 30	0.000 35	77.505	2
3	99 283	22	0.000 22	0.000 25	76.528	3
4	99 261	18	0.000 18	0.000 20	75.545	4
5	99 243	15	0.000 16	0.000 17	74.559	5
6	99 228	15	0.000 15	0.000 15	73.570	6
7	99 213	14	0.000 14	0.000 14	72.581	7
8	99 199	14	0.000 14	0.000 14	71.591	8
9	99 185	13	0.000 13	0.000 14	70.601	9
10	99 172	13	0.000 13	0.000 13	69.610	10
11	99 159	14	0.000 14	0.000 14	68.620	11
12	99 145	14	0.000 14	0.000 14	67.629	12
13	99 131	15	0.000 15	0.000 14	66.638	13
14	99 116	18	0.000 18	0.000 17	65.649	14
15	99 098	21	0.000 22	0.000 20	64.660	15
16	99 077	26	0.000 26	0.000 24	63.674	16
17	99 051	31	0.000 31	0.000 29	62.691	17
18	99 020	31	0.000 31	0.000 31	61.710	18
19	98 989	32	0.000 32	0.000 32	60.729	19
20	98 957	31	0.000 31	0.000 32	59.748	20
21	98 926	32	0.000 32	0.000 32	58.767	21
22	98 894	32	0.000 33	0.000 32	57.786	22
23	98 862	33	0.000 33	0.000 33	56.805	23
24	98 829	32	0.000 33	0.000 33	55.823	24
25	98 797	34	0.000 34	0.000 33	54.842	25
26	98 763	34	0.000 35	0.000 34	53.860	26
27	98 729	35	0.000 36	0.000 35	52.878	27
28	98 694	38	0.000 38	0.000 37	51.897	28
29	98 656	39	0.000 40	0.000 39	50.917	29
30	98 617	43	0.000 43	0.000 42	49.937	30
31	98 574	46	0.000 47	0.000 45	48.958	31
32	98 528	51	0.000 52	0.000 50	47.981	32
33	98 477	57	0.000 57	0.000 54	47.006	33
34	98 420	61	0.000 63	0.000 60	46.032	34
35	98 359	68	0.000 69	0.000 66	45.061	35
36	98 291	74	0.000 75	0.000 72	44.092	36
37	98 217	81	0.000 82	0.000 79	43.124	37
38	98 136	88	0.000 90	0.000 86	42.160	38
39	98 048	96	0.000 98	0.000 94	41.197	39
40	97 952	105	0.001 07	0.001 02	40.237	40
41	97 847	114	0.001 17	0.001 12	39.279	41
42	97 733	126	0.001 29	0.001 23	38.325	42
43	97 607	138	0.001 42	0.001 35	37.374	43
44	97 469	154	0.001 58	0.001 49	36.426	44
45	97 315	173	0.001 77	0.001 67	35.483	45
46	97 142	192	0.001 98	0.001 87	34.545	46
47	96 950	212	0.002 19	0.002 08	33.612	47
48	96 738	234	0.002 41	0.002 30	32.685	48
49	96 504	257	0.002 66	0.002 53	31.763	49
50	96 247	283	0.002 94	0.002 80	30.846	50
51	95 964	312	0.003 26	0.003 10	29.936	51
52	95 652	342	0.003 57	0.003 42	29.032	52
53	95 310	372	0.003 90	0.003 74	28.134	53
54	94 938	406	0.004 28	0.004 08	27.242	54

ELT15 (Females)

x	l_x	d_x	q_x	μ_x	$\mathring{e}_x$	x
55	94 532	450	0.004 75	0.004 51	26.357	55
56	94 082	499	0.005 31	0.005 03	25.481	56
57	93 583	554	0.005 92	0.005 62	24.614	57
58	93 029	614	0.006 60	0.006 26	23.757	58
59	92 415	683	0.007 39	0.007 00	22.912	59
60	91 732	761	0.008 30	0.007 86	22.079	60
61	90 971	839	0.009 22	0.008 80	21.259	61
62	90 132	915	0.010 15	0.009 72	20.452	62
63	89 217	1007	0.011 29	0.010 74	19.657	63
64	88 210	1117	0.012 66	0.012 03	18.875	64
65	87 093	1218	0.013 99	0.013 42	18.111	65
66	85 875	1308	0.015 23	0.014 70	17.361	66
67	84 567	1417	0.016 76	0.016 09	16.621	67
68	83 150	1533	0.018 44	0.017 74	15.896	68
69	81 617	1647	0.020 17	0.019 49	15.185	69
70	79 970	1751	0.021 90	0.021 23	14.487	70
71	78 219	1876	0.023 99	0.023 11	13.800	71
72	76 343	2056	0.026 93	0.025 69	13.127	72
73	74 287	2239	0.030 14	0.028 97	12.476	73
74	72 048	2366	0.032 84	0.032 03	11.848	74
75	69 682	2487	0.035 69	0.034 80	11.234	75
76	67 195	2634	0.039 19	0.038 03	10.631	76
77	64 561	2812	0.043 56	0.042 14	10.044	77
78	61 749	2984	0.048 33	0.046 94	9.478	78
79	58 765	3158	0.053 73	0.052 28	8.934	79
80	55 607	3314	0.059 61	0.058 27	8.413	80
81	52 293	3435	0.065 68	0.064 64	7.914	81
82	48 858	3526	0.072 16	0.071 31	7.435	82
83	45 332	3596	0.079 33	0.078 61	6.974	83
84	41 736	3655	0.087 57	0.086 91	6.532	84
85	38 081	3706	0.097 31	0.096 74	6.111	85
86	34 375	3724	0.108 33	0.108 41	5.715	86
87	30 651	3634	0.118 59	0.120 52	5.349	87
88	27 017	3475	0.128 60	0.131 74	5.002	88
89	23 542	3330	0.141 46	0.144 62	4.667	89
90	20 212	3143	0.155 50	0.160 53	4.354	90
91	17 069	2903	0.170 06	0.177 51	4.065	91
92	14 166	2631	0.185 73	0.195 73	3.797	92
93	11 535	2321	0.201 26	0.214 98	3.551	93
94	9 214	2008	0.217 90	0.234 90	3.322	94
95	7 206	1702	0.236 19	0.257 32	3.112	95
96	5 504	1395	0.253 44	0.281 14	2.925	96
97	4 109	1102	0.268 20	0.302 67	2.754	97
98	3 007	853	0.283 52	0.322 41	2.588	98
99	2 154	653	0.303 31	0.346 28	2.422	99
100	1 501	488	0.324 89	0.376 71	2.269	100
101	1 013	350	0.345 62	0.408 87	2.133	101
102	663	240	0.361 86	0.437 69	2.011	102
103	423	161	0.379 92	0.462 73	1.887	103
104	262	105	0.400 45	0.493 00	1.758	104
105	157	68	0.436 18	0.537 29	1.621	105
106	89	41	0.459 94	0.599 08	1.518	106
107	48	23	0.483 89	0.637 85	1.425	107
108	25	13	0.507 91	0.683 88	1.338	108
109	12	6	0.531 90	0.731 91	1.257	109
110	6	3	0.555 74	0.781 81	1.183	110
111	3	2	0.579 32	0.833 37	1.114	111
112	1	1	0.602 55	0.886 29	1.050	112

ELT15 (Males)

x	l_x	d_x	q_x	μ_x	$\mathring{e}_x$	x
0	100 000	814	0.008 14		73.413	0
1	99 186	62	0.000 62	0.000 80	73.019	1
2	99 124	38	0.000 38	0.000 43	72.064	2
3	99 086	30	0.000 30	0.000 33	71.091	3
4	99 056	24	0.000 24	0.000 27	70.113	4
5	99 032	22	0.000 22	0.000 23	69.130	5
6	99 010	20	0.000 20	0.000 21	68.145	6
7	98 990	18	0.000 19	0.000 19	67.158	7
8	98 972	19	0.000 18	0.000 18	66.171	8
9	98 953	18	0.000 18	0.000 18	65.183	9
10	98 935	18	0.000 18	0.000 18	64.195	10
11	98 917	18	0.000 18	0.000 18	63.206	11
12	98 899	19	0.000 19	0.000 19	62.218	12
13	98 880	23	0.000 23	0.000 21	61.230	13
14	98 857	29	0.000 29	0.000 26	60.244	14
15	98 828	39	0.000 40	0.000 34	59.261	15
16	98 789	52	0.000 52	0.000 45	58.285	16
17	98 737	74	0.000 75	0.000 64	57.315	17
18	98 663	86	0.000 87	0.000 83	56.358	18
19	98 577	81	0.000 83	0.000 85	55.406	19
20	98 496	83	0.000 84	0.000 83	54.452	20
21	98 413	85	0.000 86	0.000 85	53.497	21
22	98 328	87	0.000 89	0.000 88	52.543	22
23	98 241	87	0.000 89	0.000 89	51.589	23
24	98 154	87	0.000 88	0.000 89	50.635	24
25	98 067	84	0.000 86	0.000 87	49.679	25
26	97 983	83	0.000 85	0.000 85	48.721	26
27	97 900	83	0.000 85	0.000 84	47.762	27
28	97 817	85	0.000 87	0.000 86	46.802	28
29	97 732	87	0.000 90	0.000 88	45.842	29
30	97 645	89	0.000 91	0.000 90	44.883	30
31	97 556	91	0.000 94	0.000 92	43.923	31
32	97 465	95	0.000 97	0.000 96	42.964	32
33	97 370	97	0.000 99	0.000 98	42.005	33
34	97 273	103	0.001 06	0.001 02	41.046	34
35	97 170	113	0.001 16	0.001 11	40.090	35
36	97 057	124	0.001 27	0.001 22	39.136	36
37	96 933	133	0.001 38	0.001 33	38.185	37
38	96 800	145	0.001 49	0.001 44	37.237	38
39	96 655	155	0.001 60	0.001 55	36.292	39
40	96 500	166	0.001 72	0.001 66	35.349	40
41	96 334	179	0.001 86	0.001 79	34.409	41
42	96 155	194	0.002 01	0.001 93	33.473	42
43	95 961	210	0.002 19	0.002 10	32.539	43
44	95 751	230	0.002 40	0.002 29	31.609	44
45	95 521	255	0.002 66	0.002 53	30.684	45
46	95 266	283	0.002 97	0.002 81	29.765	46
47	94 983	315	0.003 32	0.003 14	28.852	47
48	94 668	352	0.003 71	0.003 52	27.947	48
49	94 316	391	0.004 15	0.003 93	27.049	49
50	93 925	436	0.004 64	0.004 40	26.159	50
51	93 489	485	0.005 19	0.004 92	25.279	51
52	93 004	537	0.005 77	0.005 49	24.408	52
53	92 467	594	0.006 42	0.006 10	23.547	53
54	91 873	656	0.007 14	0.006 79	22.696	54

ELT15 (Males)

x	l_x	d_x	q_x	μ_x	$\mathring{e}_x$	x
55	91 217	727	0.007 97	0.007 57	21.856	55
56	90 490	806	0.008 90	0.008 45	21.027	56
57	89 684	892	0.009 95	0.009 45	20.211	57
58	88 792	987	0.011 12	0.010 57	19.409	58
59	87 805	1 091	0.012 43	0.011 82	18.622	59
60	86 714	1 207	0.013 92	0.013 23	17.850	60
61	85 507	1 334	0.015 60	0.014 83	17.095	61
62	84 173	1 472	0.017 49	0.016 64	16.357	62
63	82 701	1 625	0.019 65	0.018 70	15.640	63
64	81 076	1 783	0.021 99	0.021 01	14.943	64
65	79 293	1 940	0.024 47	0.023 48	14.267	65
66	77 353	2 097	0.027 11	0.026 10	13.612	66
67	75 256	2 255	0.029 97	0.028 93	12.978	67
68	73 001	2 403	0.032 92	0.031 92	12.363	68
69	70 598	2 543	0.036 02	0.035 05	11.767	69
70	68 055	2 674	0.039 30	0.038 33	11.187	70
71	65 381	2 819	0.043 11	0.041 98	10.624	71
72	62 562	2 969	0.047 45	0.046 26	10.080	72
73	59 593	3 109	0.052 17	0.051 05	9.557	73
74	56 484	3 218	0.056 97	0.056 09	9.056	74
75	53 266	3 301	0.061 97	0.061 23	8.572	75
76	49 965	3 386	0.067 77	0.066 94	8.106	76
77	46 579	3 455	0.074 18	0.073 52	7.658	77
78	43 124	3 494	0.081 01	0.080 68	7.232	78
79	39 630	3 502	0.088 38	0.088 40	6.825	79
80	36 128	3 474	0.096 16	0.096 75	6.438	80
81	32 654	3 400	0.104 11	0.105 44	6.070	81
82	29 254	3 300	0.112 79	0.114 64	5.718	82
83	25 954	3 175	0.122 35	0.124 91	5.382	83
84	22 779	3 023	0.132 70	0.136 27	5.063	84
85	19 756	2 839	0.143 72	0.148 57	4.762	85
86	16 917	2 637	0.155 85	0.162 08	4.478	86
87	14 280	2 406	0.168 48	0.176 89	4.213	87
88	11 874	2 144	0.180 61	0.191 90	3.968	88
89	9 730	1 873	0.192 46	0.206 47	3.734	89
90	7 857	1 608	0.204 65	0.221 14	3.508	90
91	6 249	1 369	0.219 11	0.237 54	3.285	91
92	4 880	1 154	0.236 55	0.257 93	3.071	92
93	3 726	953	0.255 75	0.282 26	2.872	93
94	2 773	762	0.274 83	0.308 37	2.693	94
95	2 011	590	0.293 11	0.334 24	2.531	95
96	1 421	442	0.311 04	0.359 74	2.383	96
97	979	322	0.329 19	0.385 79	2.244	97
98	657	229	0.347 83	0.413 13	2.114	98
99	428	157	0.367 12	0.442 16	1.991	99
100	271	105	0.387 05	0.473 12	1.874	100
101	166	68	0.407 60	0.506 09	1.764	101
102	98	42	0.428 70	0.541 17	1.660	102
103	56	25	0.450 30	0.578 32	1.562	103
104	31	15	0.474 28	0.619 01	1.468	104
105	16	8	0.496 34	0.664 18	1.384	105
106	8	4	0.518 41	0.706 30	1.306	106
107	4	2	0.540 41	0.751 11	1.234	107
108	2	1	0.562 25	0.797 41	1.166	108
109	1	1	0.583 85	0.844 99	1.104	109

AM92

x	$l_{[x]}$	$l_{[x-1]+1}$	l_x	x
17	9 997.809 1		10 000.000 0	17
18	9 991.890 4	9 993.540 0	9 994.000 0	18
19	9 986.035 1	9 987.633 8	9 988.063 6	19
20	9 980.243 2	9 981.791 1	9 982.200 6	20
21	9 974.504 6	9 976.001 6	9 976.390 9	21
22	9 968.839 1	9 970.265 4	9 970.634 6	22
23	9 963.196 7	9 964.582 4	9 964.931 3	23
24	9 957.577 5	9 958.922 5	9 959.261 3	24
25	9 951.991 3	9 953.285 8	9 953.614 4	25
26	9 946.398 2	9 947.662 2	9 947.980 7	26
27	9 940.798 4	9 942.021 8	9 942.340 2	27
28	9 935.181 8	9 936.354 9	9 936.673 0	28
29	9 929.508 8	9 930.661 3	9 930.969 4	29
30	9 923.749 7	9 924.891 6	9 925.209 4	30
31	9 917.914 5	9 919.026 0	9 919.353 5	31
32	9 911.953 8	9 913.054 7	9 913.382 1	32
33	9 905.828 2	9 906.928 5	9 907.265 5	33
34	9 899.498 4	9 900.607 8	9 900.964 5	34
35	9 892.915 1	9 894.053 6	9 894.429 9	35
36	9 886.039 5	9 887.206 9	9 887.612 6	36
37	9 878.812 8	9 880.028 8	9 880.454 0	37
38	9 871.166 5	9 872.450 8	9 872.895 4	38
39	9 863.022 7	9 864.404 7	9 864.868 8	39
40	9 854.303 6	9 855.793 1	9 856.286 3	40
41	9 844.902 5	9 846.538 4	9 847.051 0	41
42	9 834.703 0	9 836.524 5	9 837.066 1	42
43	9 823.599 4	9 825.635 4	9 826.206 0	43
44	9 811.447 3	9 813.746 3	9 814.335 9	44
45	9 798.083 7	9 800.693 9	9 801.312 3	45
46	9 783.337 1	9 786.316 2	9 786.953 4	46
47	9 766.998 3	9 770.423 1	9 771.078 9	47
48	9 748.860 3	9 752.787 4	9 753.471 4	48
49	9 728.649 9	9 733.193 8	9 733.886 5	49
50	9 706.097 7	9 711.352 4	9 712.072 8	50
51	9 680.899 0	9 686.966 9	9 687.714 9	51
52	9 652.696 5	9 659.707 5	9 660.502 1	52
53	9 621.100 6	9 629.211 5	9 630.052 2	53
54	9 585.691 6	9 595.056 3	9 595.971 5	54
55	9 545.992 9	9 556.800 3	9 557.817 9	55
56	9 501.483 9	9 513.937 5	9 515.104 0	56
57	9 451.593 8	9 465.929 3	9 467.290 6	57
58	9 395.697 1	9 412.171 2	9 413.800 4	58
59	9 333.128 4	9 352.016 5	9 354.004 0	59
60	9 263.142 2	9 284.764 1	9 287.216 4	60
61	9 184.968 7	9 209.656 8	9 212.714 3	61
62	9 097.740 5	9 125.881 8	9 129.717 0	62
63	9 000.588 4	9 032.564 2	9 037.397 3	63
64	8 892.574 1	8 928.817 7	8 934.877 1	64

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x	$l_{[x]}$	$l_{[x-1]+1}$	l_x	x
65	8 772.735 9	8 813.688 1	8 821.261 2	65
66	8 640.048 1	8 686.201 6	8 695.619 9	66
67	8 493.518 7	8 545.353 2	8 557.011 8	67
68	8 332.139 6	8 390.161 1	8 404.491 6	68
69	8 154.931 8	8 219.639 0	8 237.132 9	69
70	7 960.977 6	8 032.860 6	8 054.054 4	70
71	7 749.465 9	7 828.968 6	7 854.450 8	71
72	7 519.702 7	7 607.240 0	7 637.620 8	72
73	7 271.146 1	7 367.082 8	7 403.008 4	73
74	7 003.521 6	7 108.105 2	7 150.240 1	74
75	6 716.823 1	6 830.184 4	6 879.167 3	75
76	6 411.345 9	6 533.500 8	6 589.925 8	76
77	6 087.808 4	6 218.575 9	6 282.980 3	77
78	5 747.362 4	5 886.362 8	5 959.168 0	78
79	5 391.640 0	5 538.279 1	5 619.757 7	79
80	5 022.793 1	5 176.222 4	5 266.460 4	80
81	4 643.512 9	4 802.629 0	4 901.478 9	81
82	4 257.005 6	4 420.452 5	4 527.496 0	82
83	3 866.988 4	4 033.146 7	4 147.670 8	83
84	3 477.592 9	3 644.632 7	3 765.599 8	84
85	3 093.286 3	3 259.186 2	3 385.247 9	85
86	2 718.712 8	2 881.346 7	3 010.839 5	86
87	2 358.529 9	2 515.731 0	2 646.741 6	87
88	2 017.229 8	2 166.880 5	2 297.297 6	88
89	1 698.908 9	1 839.045 8	1 966.649 9	89
90	1 407.055 0	1 535.980 1	1 658.554 5	90
91		1 260.735 4	1 376.190 6	91
92			1 121.988 9	92
93			897.502 5	93
94			703.324 2	94
95			539.064 3	95
96			403.402 3	96
97			294.206 1	97
98			208.706 0	98
99			143.712 0	99
100			95.847 6	100
101			61.773 3	101
102			38.379 6	102
103			22.928 4	103
104			13.135 9	104
105			7.196 8	105
106			3.759 6	106
107			1.866 9	107
108			0.878 4	108
109			0.390 3	109
110			0.163 2	110
111			0.064 0	111
112			0.023 4	112
113			0.008 0	113
114			0.002 5	114
115			0.000 7	115
116			0.000 2	116
117			0.000 0	117
118			0.000 0	118
119			0.000 0	119
120			0.000 0	120

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x	l_x	d_x	q_x	μ_x	$\mathring{e}_x$	x
50	9 941.923	5.418	0.000 545	0.000 507	34.10	50
51	9 936.504	6.260	0.000 630	0.000 585	33.12	51
52	9 930.244	7.249	0.000 730	0.000 677	32.14	52
53	9 922.995	8.415	0.000 848	0.000 786	31.17	53
54	9 914.580	9.776	0.000 986	0.000 914	30.19	54
55	9 904.805	11.371	0.001 148	0.001 063	29.22	55
56	9 893.434	13.237	0.001 338	0.001 239	28.25	56
57	9 880.196	15.393	0.001 558	0.001 444	27.29	57
58	9 864.803	17.895	0.001 814	0.001 681	26.33	58
59	9 846.908	20.777	0.002 110	0.001 957	25.38	59
60	9 826.131	24.084	0.002 451	0.002 266	24.43	60
61	9 802.048	28.965	0.002 955	0.002 685	23.49	61
62	9 773.083	34.694	0.003 550	0.003 241	22.56	62
63	9 738.388	41.398	0.004 251	0.003 889	21.64	63
64	9 696.990	49.193	0.005 073	0.004 651	20.73	64
65	9 647.797	58.195	0.006 032	0.005 543	19.83	65
66	9 589.602	68.537	0.007 147	0.006 583	18.95	66
67	9 521.065	80.348	0.008 439	0.007 792	18.08	67
68	9 440.717	93.746	0.009 930	0.009 191	17.23	68
69	9 346.970	108.836	0.011 644	0.010 806	16.40	69
70	9 238.134	125.685	0.013 605	0.012 661	15.59	70
71	9 112.449	144.350	0.015 841	0.014 783	14.79	71
72	8 968.099	164.834	0.018 380	0.017 204	14.02	72
73	8 803.265	187.096	0.021 253	0.019 956	13.28	73
74	8 616.170	211.010	0.024 490	0.023 072	12.55	74
75	8 405.160	236.362	0.028 121	0.026 587	11.86	75
76	8 168.798	262.864	0.032 179	0.030 537	11.18	76
77	7 905.934	290.116	0.036 696	0.034 962	10.54	77
78	7 615.818	317.595	0.041 702	0.039 899	9.92	78
79	7 298.223	344.688	0.047 229	0.045 390	9.33	79
80	6 953.536	370.644	0.053 303	0.051 473	8.77	80
81	6 582.891	394.658	0.059 952	0.058 188	8.23	81
82	6 188.234	415.856	0.067 201	0.065 576	7.73	82
83	5 772.378	433.321	0.075 068	0.073 676	7.25	83
84	5 339.057	446.180	0.083 569	0.082 522	6.80	84
85	4 892.878	453.648	0.092 716	0.092 149	6.37	85
86	4 439.230	455.092	0.102 516	0.102 590	5.97	86
87	3 984.138	450.084	0.112 969	0.113 873	5.59	87
88	3 534.054	438.463	0.124 068	0.126 023	5.24	88
89	3 095.591	420.387	0.135 802	0.139 060	4.91	89
90	2 675.203	396.334	0.148 151	0.152 998	4.61	90
91	2 278.869	367.099	0.161 088	0.167 846	4.32	91
92	1 911.771	333.759	0.174 581	0.183 606	4.06	92
93	1 578.012	297.596	0.188 589	0.200 273	3.81	93
94	1 280.416	260.008	0.203 065	0.217 836	3.59	94
95	1 020.409	222.405	0.217 957	0.236 273	3.38	95
96	798.003	186.098	0.233 205	0.255 556	3.18	96
97	611.905	152.209	0.248 746	0.275 647	3.00	97
98	459.696	121.595	0.264 511	0.296 499	2.84	98
99	338.101	94.813	0.280 429	0.318 054	2.68	99
100	243.288	72.117	0.296 425	0.340 247	2.54	100
101	171.171	53.478	0.312 423	0.363 002	2.41	101
102	117.693	38.644	0.328 344	0.386 232	2.29	102
103	79.050	27.202	0.344 113	0.409 842	2.18	103
104	51.848	18.647	0.359 653	0.433 729	2.08	104
105	33.200	12.446	0.374 887	0.457 778	1.99	105

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x	l_x	d_x	q_x	μ_x	$\mathring{e}_x$	x
50	9 952.697	5.245	0.000 527	0.000 492	37.08	50
51	9 947.452	5.998	0.000 603	0.000 563	36.10	51
52	9 941.454	6.879	0.000 692	0.000 645	35.12	52
53	9 934.574	7.898	0.000 795	0.000 741	34.15	53
54	9 926.676	9.053	0.000 912	0.000 851	33.17	54
55	9 917.623	10.374	0.001 046	0.000 976	32.20	55
56	9 907.249	11.879	0.001 199	0.001 120	31.24	56
57	9 895.370	13.606	0.001 375	0.001 284	30.27	57
58	9 881.764	15.564	0.001 575	0.001 472	29.31	58
59	9 866.200	17.769	0.001 801	0.001 685	28.36	59
60	9 848.431	20.268	0.002 058	0.001 918	27.41	60
61	9 828.163	23.991	0.002 441	0.002 236	26.46	61
62	9 804.173	28.285	0.002 885	0.002 655	25.53	62
63	9 775.888	33.248	0.003 401	0.003 135	24.60	63
64	9 742.640	38.932	0.003 996	0.003 691	23.68	64
65	9 703.708	45.423	0.004 681	0.004 332	22.78	65
66	9 658.285	52.802	0.005 467	0.005 069	21.88	66
67	9 605.483	61.158	0.006 367	0.005 914	21.00	67
68	9 544.325	70.580	0.007 395	0.006 882	20.13	68
69	9 473.745	81.124	0.008 563	0.007 986	19.28	69
70	9 392.621	92.874	0.009 888	0.009 240	18.44	70
71	9 299.747	105.887	0.011 386	0.010 663	17.62	71
72	9 193.860	120.210	0.013 075	0.012 272	16.81	72
73	9 073.650	135.860	0.014 973	0.014 086	16.03	73
74	8 937.791	152.836	0.017 100	0.016 126	15.27	74
75	8 784.955	171.113	0.019 478	0.018 414	14.52	75
76	8 613.841	190.598	0.022 127	0.020 974	13.80	76
77	8 423.243	211.162	0.025 069	0.023 829	13.10	77
78	8 212.080	232.615	0.028 326	0.027 004	12.42	78
79	7 979.465	254.729	0.031 923	0.030 527	11.77	79
80	7 724.737	277.179	0.035 882	0.034 425	11.14	80
81	7 447.558	299.593	0.040 227	0.038 728	10.54	81
82	7 147.965	321.523	0.044 981	0.043 464	9.96	82
83	6 826.442	342.455	0.050 166	0.048 664	9.41	83
84	6 483.987	361.832	0.055 804	0.054 357	8.88	84
85	6 122.154	379.053	0.061 915	0.060 576	8.37	85
86	5 743.101	393.506	0.068 518	0.067 349	7.89	86
87	5 349.595	404.595	0.075 631	0.074 708	7.43	87
88	4 945.000	411.770	0.083 270	0.082 686	7.00	88
89	4 533.230	414.537	0.091 444	0.091 308	6.59	89
90	4 118.693	412.545	0.100 164	0.100 604	6.20	90
91	3 706.149	405.590	0.109 437	0.110 601	5.84	91
92	3 300.559	393.644	0.119 266	0.121 325	5.49	92
93	2 906.914	376.882	0.129 650	0.132 801	5.17	93
94	2 530.033	355.677	0.140 582	0.145 048	4.87	94
95	2 174.356	330.617	0.152 053	0.158 084	4.58	95
96	1 843.738	302.467	0.164 051	0.171 926	4.32	96
97	1 541.271	272.119	0.176 555	0.186 586	4.07	97
98	1 269.152	240.562	0.189 545	0.202 071	3.84	98
99	1 028.591	208.795	0.202 991	0.218 386	3.62	99
100	819.796	177.783	0.216 863	0.235 531	3.41	100
101	642.013	148.385	0.231 125	0.253 502	3.22	101
102	493.627	121.303	0.245 737	0.272 288	3.05	102
103	372.325	97.048	0.260 654	0.291 872	2.89	103
104	275.277	75.930	0.275 830	0.312 234	2.73	104
105	199.347	58.053	0.291 217	0.333 348	2.59	105